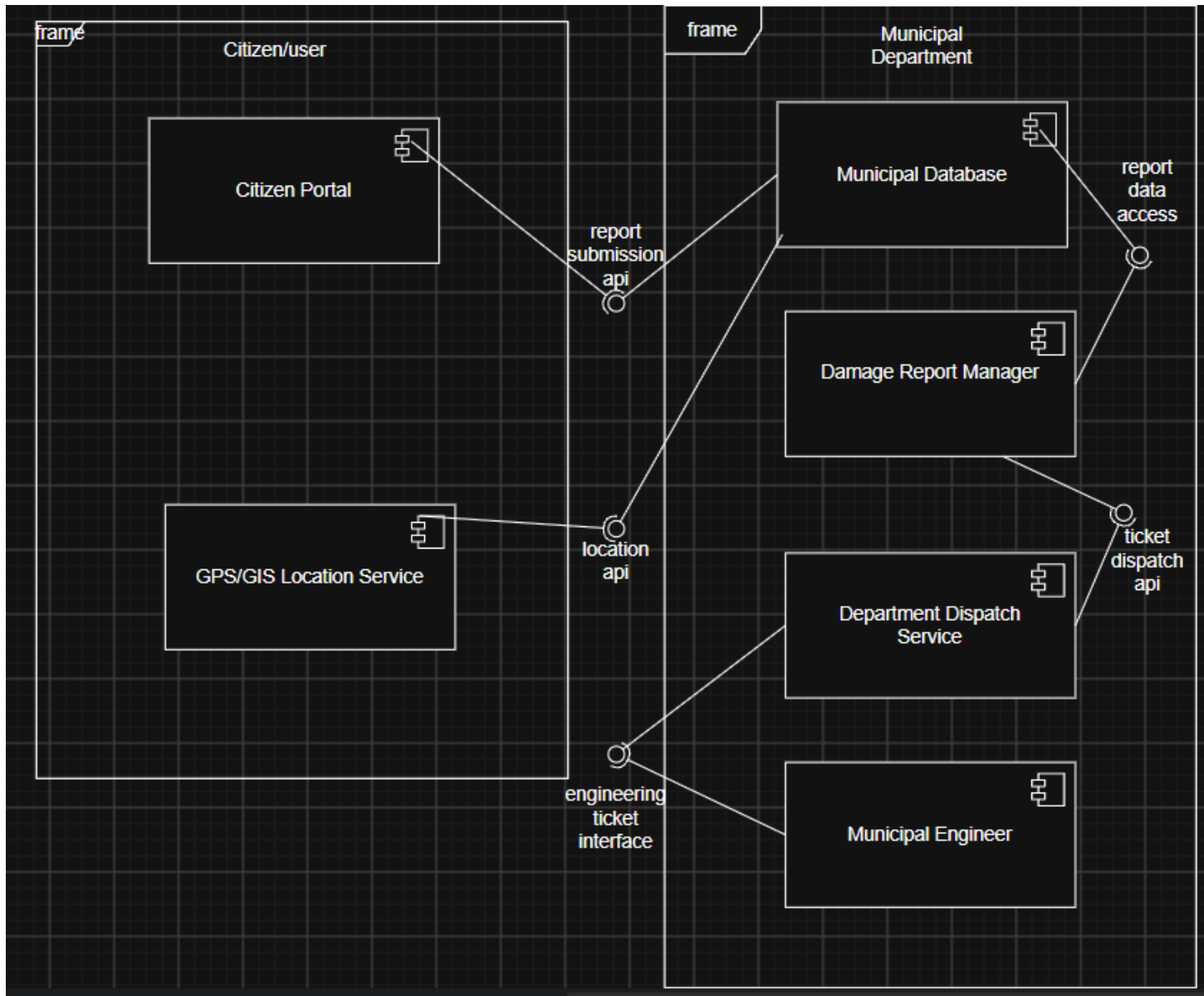


SAMARTH P KULKARNI

PES1UG24AM464

SE_LAB3

DELIVERABLE 1:



DELIVERABLE 2:

Architecture Selection

I chose **Layered Architecture** for the Municipal Infrastructure Damage Reporting App.

Architectural Choice:

The system is divided into presentation, business logic, and data/integration layers. The Citizen Portal forms the presentation layer, while the Damage Report Manager handles the core business logic. The Municipal Database and external services such as the GPS/GIS Location Service and Department Dispatch Service support data storage and system integration.

Reason 1 – Separation of Concerns:

Layered architecture separates citizen interaction from the processing of damage reports. The Citizen Portal is responsible for collecting information, while the Damage Report Manager validates reports, extracts location information, creates tickets, and manages their status. This makes the system easier to understand and maintain.

Reason 2 – Suitable for the Application:

The application follows a relatively clear processing flow: a citizen submits a damage report, the system determines its location and type, stores the report, and dispatches it to the appropriate municipal department. A layered structure is therefore suitable because each part of this workflow can be handled by a dedicated component.

Security Advantage:

The architecture prevents the Citizen Portal from directly accessing the Municipal Database or internal municipal services. Requests can be validated by the Damage Report Manager before information is stored or dispatched, providing a controlled point for authentication, authorization, and input validation.

Performance Benefit:

The system can process frequently used operations through dedicated services without requiring every component to perform all tasks. For example, the GPS/GIS service can independently handle location extraction while the report manager coordinates the remaining processing. This reduces unnecessary processing in the citizen-facing interface and helps provide faster responses.